SPARKING: A Frozen Study of Support-Gated Residual Learning for Composition-Only Band-Gap Prediction

Ryan Gomez

October 20, 2026

Abstract

Composition-only materials models are inexpensive screening surrogates, but their accuracy under chemical support shift is easy to overstate. We specify SPARKING, a small mechanism that decomposes a prediction into a regularized linear term and a nonlinear residual correction, then attenuates only the residual as standardized nearest-neighbour distance exceeds a training-derived support threshold. The central, falsifiable hypothesis is that this gate lowers leave-one-element-out mean absolute error (MAE) relative to an otherwise identical ungated residual model without degrading official Matbench MAE by more than 0.02 eV. Development is limited to eight gates over Cu, Ag, and B; Li, Zn, Ga, Se, and Te are protected confirmatory scenarios. The implemented system hash-binds the 4,604-row Matbench experimental-gap artifact and official folds, enforces a 512 MiB/no-rising-pageout resource contract, retains failed runs, and provides a compute-matched mechanism-off control, strong simple baselines, marginal split-conformal intervals, and claim-evidence manifests. The first local development attempt was validly refused before fitting because page-outs increased by eight during featurization; therefore no predictive or positive scientific result is claimed in this pre-confirmatory execution record. The study becomes outcome-bearing only after a hosted development run, a separate frozen confirmatory protocol, and protected execution.

1 Introduction

Fast property surrogates can reduce the cost of triaging large materials spaces, but a composition alone omits crystal structure, defects, process-

ing, and measurement context. Matbench formalized reproducible materials-learning tasks and official folds (Dunn et al., 2020). Hand-engineered composition descriptors (Ward et al., 2016), stoichiometric graph models such as Roost (Goodall and Lee, 2020), and attention models such as CrabNet (Wang et al., 2021) show that useful prediction is possible without a known structure. That utility does not imply reliable extrapolation.

Random or redundant splits can reward interpolation between similar materials. Materials-domain analyses have consequently emphasized domains of applicability (Sutton et al., 2020), and explicit redundancy control can materially lower reported performance while better exposing extrapolative difficulty (Li et al., 2024). The practical failure is not merely that a query is far from training data; it is that a flexible nonlinear correction can remain large when its empirical support is weak.

We study a deliberately minimal response. A ridge component supplies a smooth, regularized extrapolator over elemental fractions and periodic moments. A nonlinear learner fits only the ridge residual. A deterministic support gate is one inside a training-derived distance threshold and decays outside it. This design does not assert that linear extrapolation is physically correct. It tests the narrower proposition that withdrawing an unsupported nonlinear correction is safer than applying it unchanged.

The contribution is threefold: (i) the precise support-gated residual mechanism; (ii) a compute-matched mechanism-off comparison under pre-registered element exclusions and official folds; and (iii) an evidence system that prevents partial, resource-refused, or unverified runs from becoming positive claims.

2 Related Work and Novelty Boundary

Magpie-style models summarize stoichiometry with elemental statistics and train general-purpose regressors (Ward et al., 2016). Roost instead treats a composition as a weighted element graph and learns contextual representations (Goodall and Lee, 2020). CrabNet uses self-attention over element-derived representations and predicts element contributions and uncertainty (Wang et al., 2021). These models motivate composition-only learning but are not reproduced here as outcome-bearing controls: the central question concerns the incremental gate under strict parameter and compute matching, not a claim of state-of-the-art IID accuracy.

Domain-of-applicability methods characterize regions in which model errors are lower (Sutton et al., 2020). MD-HIT constructs redundancy-

controlled datasets and demonstrates that performance commonly deteriorates as similarity to training samples falls (Li et al., 2024). SPARKING differs by using training-only distance to intervene on a residual correction, while evaluating that intervention against the same residual model with the intervention off. Nearest-neighbour distance, residual learning, ridge regression, gradient boosting, and conformal prediction are not claimed as new. The novelty claim is restricted to this explicit composition-space mechanism and its frozen test.

Split conformal inference gives finite-sample marginal coverage under exchangeability (Lei et al., 2018); exact distribution-free conditional coverage is generally unavailable without assumptions (Barber et al., 2021). Accordingly, our 90% intervals are compatibility diagnostics on official outer folds, not guarantees under element shift.

3 Problem Formulation

Let a composition c be a multiset of elements e and positive stoichiometric counts a_e. Fractions are

$$f_e(c) = \frac{a_e}{\sum_j a_j}, \qquad \sum_e f_e(c) = 1. \tag{1}$$

The target $y \in \mathbb{R}_{\geq 0}$ is experimental band gap in eV. A model $h : \mathcal{C} \to \mathbb{R}_{\geq 0}$ is trained on a set $D = \{(c_i, y_i)\}_{i=1}^{n}$. The principal loss is absolute error $\ell(y, h(c)) = |y - h(c)|$.

For an element E, the leave-one-element-out training set contains every row whose composition excludes E; the test set contains every row including E. The primary estimand for element E is the paired MAE improvement

$$\Delta_E = \mathrm{MAE}(h_{\mathrm{ungated}}; E) - \mathrm{MAE}(h_{\textsc{Sparking}}; E). \tag{2}$$

The confirmatory primary endpoint is $\bar{\Delta}$, the unweighted mean over the five protected elements. The compatibility estimand is the pooled official five-fold MAE difference. These estimands do not identify a causal effect in a materials population.

4 Model

4.1 Fixed composition representation

The descriptor $x(c) \in \mathbb{R}^{132}$ concatenates 118 elemental fractions with 14 global features: number of elements; fraction entropy, minimum, maximum,

and ℓ_2 norm; weighted mean and standard deviation of atomic number, period, and group; minimum and maximum atomic number; and group range. Formulae are parsed with nested-parenthesis and fractional-stoichiometry support. Invalid syntax is rejected rather than repaired.

4.2 Linear component

Each feature is standardized with training mean μ_j and scale s_j. Ridge selects λ from 17 logarithmically spaced values in $[10^{-4}, 10^4]$ by its internal generalized cross-validation criterion and minimizes

$$\hat{\beta}_\lambda = \arg\min_\beta \sum_{i=1}^{n} (y_i - \beta_0 - z_i^\top \beta)^2 + \lambda \|\beta\|_2^2, \quad z_{ij} = \frac{x_{ij} - \mu_j}{s_j}. \tag{3}$$

Its prediction is $b(x) = \beta_0 + z^\top \hat{\beta}$.

4.3 Nonlinear residual

Training residuals are $r_i = y_i - b(x_i)$. A histogram gradient-boosting regressor $g(x)$ minimizes absolute-error loss using 180 boosting iterations, learning rate 0.055, at most 31 leaves per tree, minimum leaf size 12, and ℓ_2 regularization 0.1. Early stopping is disabled and the random seed is 20261020. The mechanism-off control is

$$h_{\text{ungated}}(x) = \max\{0, b(x) + g(x)\}. \tag{4}$$

4.4 Training-support gate

A second training-fitted standardizer maps descriptors to $u(x)$. Let d_i be the Euclidean distance from training point i to its nearest other training point, and let τ_q be the empirical q-quantile of $\{d_i\}$. For a query, $d(x) = \min_i \|u(x) - u(x_i)\|_2$. The gate is

$$w_{q,p}(x) = \left(\frac{\tau_q}{\max\{d(x), \tau_q\}} \right)^p, \tag{5}$$

and the final prediction is

$$h_{q,p}(x) = \max\{0, b(x) + w_{q,p}(x) g(x)\}. \tag{6}$$

Thus $w = 1$ for in-threshold queries, $0 < w < 1$ outside, and no additional predictive parameter is learned by the gate.

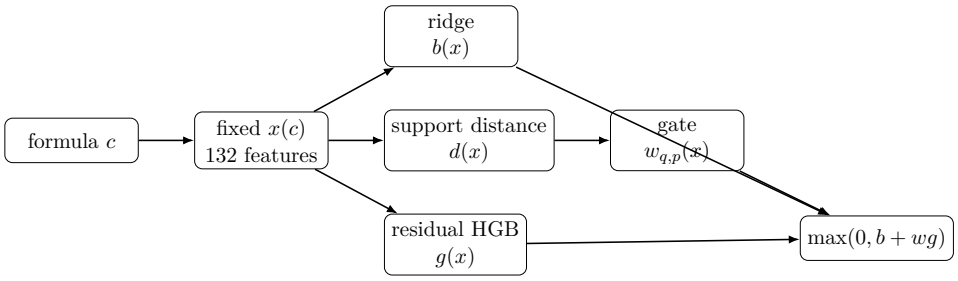


Figure 1: The gate acts only on the nonlinear residual. The ungated control fixes $w = 1$ while retaining the same fitted ridge and residual models.

Algorithm 1 Frozen development selection

for $E \in \{\mathrm{Cu}, \mathrm{Ag}, \mathrm{B}\}$ **do**
    exclude every composition containing E from training
    fit ridge and one residual learner
    for $q \in \{0.80, 0.90, 0.95, 0.99\}$ and $p \in \{1, 2\}$ **do**
        evaluate $h_{q,p}$ on all compositions containing E
    end for
    evaluate median, ridge, direct HGB, and ungated residual controls
end for
select the minimum macro-MAE gate
for candidates within 0.002 eV, choose lower p, then higher q
emit a hash-verified freeze candidate; do not open confirmation

4.5 Complexity and failure behavior

For n rows, $d = 132$ features, and T boosting iterations, ridge fitting is bounded by the library solver, histogram boosting is approximately $O(Tnd)$ aside from tree overhead, and brute-force support fitting/querying is $O(n^2 d)/O(nmd)$ for m queries at this dataset scale. Any non-finite feature, distance, or prediction is fatal. Model outputs are clipped only at the physical lower bound zero; no upper target bound is imposed.

5 Experimental Methodology

5.1 Dataset and integrity

We use Matbench v0.1 `matbench_expt_gap`, 4,604 compositions with experimental band gaps from 0.0 to 11.7 eV. The compressed artifact is accepted

only at SHA-256 `783e7d1461eb83b00b2f2942da4b95fda5e58a0d1ae26b581c24cf8a82ca75b2`. Local scale/order canonicalization yields 4,601 unique compositions. The official five-fold split is regenerated with the declared Matbench seed and must match a separately retained canonical digest. Official folds contain known canonical overlap in folds 0–3 and are not treated as OOD evidence.

5.2 Controls and endpoints

Controls are a training-target median, ridge, a direct HGB model, and the compute-matched ungated residual. Development sees Cu, Ag, and B only. The protected confirmatory elements are Li, Zn, Ga, Se, and Te. Confirmation is supported only if $\bar{\Delta} > 0.01$ eV, at least four of five element effects are positive, and official pooled MAE is no more than 0.02 eV worse than the ungated control. We report per-element MAE, RMSE, median absolute error, bias, gate statistics, runtime, peak RSS, and pageouts.

5.3 Uncertainty and statistics

For each official fold, a separate 80/20 split of the outer-training rows forms proper training and calibration sets. If calibration absolute residuals sorted as $s_{(1)} \le \cdots \le s_{(n_c)}$, the 90% split-conformal radius is

$$q_{0.9} = s_{(k)}, \qquad k = \min\{n_c, \lceil (n_c + 1)0.9 \rceil\}. \tag{7}$$

Intervals are $[\max(0, \hat{y} - q_{0.9}), \hat{y} + q_{0.9}]$. We report observed coverage, mean width, and interval score. These are marginal IID diagnostics.

The five element improvements are reported individually, with a 10,000-resample bootstrap over elements and an exact one-sided sign-permutation statistic. The small, overlapping scenario set precludes strong asymptotic claims; the practical-effect and sign-count rule is primary.

5.4 Resource, provenance, and stopping rules

Each run must begin and remain below 512 MiB peak process RSS, with no increase in macOS `Pageouts` or Linux `pswpout`. Every phase is checked. A refusal writes a failure trace, resource observations, receipt, and a SHA-256 manifest, then stops without weakening the gate. Run IDs are immutable. The development budget ends after eight gates; protected confirmation runs once under a separately frozen protocol.

6 Implementation Verification

The package exposes `prepare`, `preflight`, `develop`, `confirm`, and `verify`. Formula tests cover nested groups, fractional stoichiometry, scaling invariance, and malformed input. Mathematical tests hand-check regression and interval metrics. Split tests regenerate all fold sizes, prove train/test disjointness and completeness, and match the canonical split digest. Model tests show that in-support gates equal one, far-out gates decay, and the mechanism-off prediction equals $b + g$. Manifest tests substitute and add files to ensure verification fails closed.

The isolated Python 3.12 environment resolves 29 packages. Twelve tests pass; the source distribution and wheel build successfully. This establishes software behavior, not scientific performance.

7 Execution Results

Table 1 is the complete outcome-bearing record at this stage. No fold, element scenario, predictive metric, or interval result exists.

Table 1: Execution receipts before hosted development.

Run or gate	Status	Verified observation
Dataset preparation	PASS	4,604 rows; source and official-split digests match
Software suite	PASS	12 tests; package and wheel build
Local preflight	ADMITTED	124.5 MiB peak RSS; pageout delta 0
Local development v1-r1	FAILED	retained after features; 129.3 MiB peak RSS; pageouts +8
Hosted development	NOT EXECUTED	explicit source-export authorization required
Confirmatory study	NOT AUTHORIZED	no development freeze candidate exists

The local failure occurred before fitting any model. It cannot support or refute the mechanism, and it supplies no basis for selecting a gate. The evidence does support a narrower engineering conclusion: the feature path fits well below the RSS ceiling, but this macOS host does not satisfy the no-rising-pageout condition even for the bounded preparation phase. This

reproduces the predecessor's resource limitation rather than curing it by relaxing the rule.

8 Failure Analysis and Threats to Validity

Representation. Atomic-number, period, and group moments are coarse; distance in this space is neither chemical equivalence nor an error probability.

Identifiability. Composition-only inputs conflate polymorphs and experimental contexts, imposing irreducible ambiguity.

Selection. Eight gate candidates may overfit three development elements; confirmation is therefore separated and immutable.

Dependence. Element test sets overlap, so five effects are not five independent samples from a well-defined population.

Benchmark leakage. Canonical duplicates across official folds can optimistically bias IID compatibility metrics.

Baselines. The controls isolate the gate but do not establish state-of-the-art performance against every deep composition model. A positive mechanism result would be an ablation result, not an overall leaderboard claim.

Conformal scope. Marginal coverage relies on exchangeability and is not promised under the protected element shift.

9 Discussion

The study design makes a narrow improvement hard to fake. Capacity and training compute are matched because the gate operates after the same ridge and residual models are fitted. IID and shifted endpoints are separated. The development grid and tie-break are fixed. A valid negative outcome would show that support distance does not reliably identify harmful residual corrections in this representation; a mixed outcome would motivate studying which chemical regimes make the linear fallback safer, but only in a separately versioned successor.

Even a supported result would not make SPARKING a materials-discovery engine. It would establish that one conservative intervention improved one composition-only extrapolation protocol while preserving one public benchmark within tolerance. Candidate stability, synthesis, novelty, and utility would remain untested.

10 Conclusion

We have specified and implemented a fully matched test of support-gated residual learning for experimental band gaps. The data, splits, model, development budget, protected elements, statistical rules, and failure behavior are frozen. Software and data-integrity gates pass, while the first numerical attempt was resource-refused before model fitting. Consequently, the scientifically correct current disposition is `NOT_EVALUATED`, not a favorable result. Hosted development, confirmatory freezing, and one protected execution are the exact remaining steps before the central question becomes answerable.

Reproducibility Statement

The repository contains the canonical research state, machine-readable protocol, dataset and model cards, pinned lockfile, exact source and split digests, test suite, package build configuration, hosted runner, failed local receipt, and artifact verifier. A hosted run must emit its environment, source revision, resource trace, raw per-condition metrics, model checkpoints, receipt, and file manifest. Numerical manuscript text may be regenerated only from a successful, manifest-valid confirmatory bundle.

References

Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. The limits of distribution-free conditional predictive inference. *Information and Inference*, 10(2):455–482, 2021. doi: 10.1093/imaiai/iaaa017.

Alexander Dunn, Qi Wang, Alex Ganose, Daniel Dopp, and Anubhav Jain. Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. *npj Computational Materials*, 6:138, 2020. doi: 10.1038/s41524-020-00406-3.

Rhys E. A. Goodall and Alpha A. Lee. Predicting materials properties without crystal structure: deep representation learning from stoichiometry. *Nature Communications*, 11:6280, 2020. doi: 10.1038/s41467-020-19964-7.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018. doi: 10.1080/01621459.2017.1307116.

Qin Li, Nihang Fu, Sadman Sadeed Omee, and Jianjun Hu. MD-HIT: Machine learning for material property prediction with dataset redundancy control. *npj Computational Materials*, 10:245, 2024. doi: 10.1038/s41524-024-01426-z.

Christopher Sutton, Mario Boley, Luca M. Ghiringhelli, Matthias Rupp, Jilles Vreeken, and Matthias Scheffler. Identifying domains of applicability of machine learning models for materials science. *Nature Communications*, 11, 2020. doi: 10.1038/s41467-020-17112-9.

Anthony Yu-Tung Wang, Steven K. Kauwe, Ryan J. Murdock, and Taylor D. Sparks. Compositionally restricted attention-based network for materials property predictions. *npj Computational Materials*, 7:77, 2021. doi: 10.1038/s41524-021-00545-1.

Logan Ward, Ankit Agrawal, Alok Choudhary, and Christopher Wolverton. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Computational Materials*, 2:16028, 2016. doi: 10.1038/npjcompumats.2016.28.

A Chronological Forward, Training, and Inference Processes

At training time the parser produces normalized fractions, the featurizer emits 132 finite values, ridge fits the target, HGB fits ridge residuals, the support standardizer and neighbour index fit training descriptors, and the training leave-one-out distances determine τ_q. At inference the same parser and feature order are used; ridge, residual, nearest distance, gate, and non-negative projection are applied in that order. No test label enters any transformation.

B Claim-Evidence Matrix

Claim	Required artifact	Current state
Input and folds are pinned	dataset receipt and split digest	supported
Implementation follows equations	tests and source	supported as software
Gate improves element exclusion	protected element metrics	not evaluated
Gate preserves official MAE	official OOF prediction table	not evaluated
90% intervals calibrate	fold interval metrics	not evaluated
Method discovers materials	structural/experimental evidence	prohibited

C Exact Reproduction Commands

```
UV_CACHE_DIR=/tmp/sparking-uv-cache uv sync --extra dev --locked
UV_CACHE_DIR=/tmp/sparking-uv-cache uv run sparking prepare
UV_CACHE_DIR=/tmp/sparking-uv-cache uv run pytest -q
UV_CACHE_DIR=/tmp/sparking-uv-cache uv run sparking preflight
UV_CACHE_DIR=/tmp/sparking-uv-cache uv run sparking develop \
  --run-id development-v1
UV_CACHE_DIR=/tmp/sparking-uv-cache uv run sparking verify \
  --run-id development-v1
```